Reinforcement Learning · Lecture 4

Self-Predictive Representations

1 Setup

Every method has this shape:

$$\mathbf{z}_t = f_\theta(\mathbf{o}_t), \qquad \hat{\mathbf{z}}_{t+1} = g_\phi(\mathbf{z}_t, \mathbf{a}_t), \qquad \mathcal{L}_{\text{pred}} = \mathbb{E}\left\|\hat{\mathbf{z}}_{t+1} - \text{sg}(f_{\theta'}(\mathbf{o}_{t+1}))\right\|^2.$$

- Problem: $f_\theta \equiv \text{const}$ drives the loss to zero, i.e. **collapse**.
- Fixes: stop-grad, redundancy reduction, distribution matching.
  - SIGReg pushes latents toward $\mathcal{N}(\mathbf{0}, \mathbf{I})$.

Definition 1.1 (Collapse). An encoder f *collapses* if $\text{Var}[f(\mathbf{o})] = 0$.

Theorem 1.2 (Cramér–Wold). A random vector $\mathbf{z}$ is $\mathcal{N}(\mathbf{0}, \mathbf{I})$ iff $\mathbf{u}^\top \mathbf{z} \sim \mathcal{N}(0, 1)$ for every unit $\mathbf{u}$.

The proof is in Section A.[1]

Symbol	Meaning
$\mathbf{o}_t$, $\mathbf{a}_t$	observation and action at time t
$\mathbf{z}_t = f_\theta(\mathbf{o}_t)$	latent state
$\text{sg}(\cdot)$	stop-gradient

Algorithm 1 SIGReg training step

Input: batch $\{\mathbf{o}_i\}$, directions K, weight λ

1: $\mathbf{Z} \leftarrow f_\theta(\mathbf{o})$
2: $\mathcal{L} \leftarrow \mathcal{L}_{\text{pred}} + \lambda\,\text{SIGReg}(\mathbf{Z})$
3: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$

Algorithm 1 uses Theorem 1.2.

Remark 1.3. SIGReg removes the need for an EMA target. 中文內容也可以直接混排。

sigreg.py

```python
def sigreg(z, num_dirs=64):
    u = torch.randn(z.shape[1], num_dirs)   # random directions
    u = u / u.norm(dim=0, keepdim=True)
    return epps_pulley(z @ u).mean()
```

Table 1: Five anti-collapse mechanisms.

Mechanism	Examples
Stop-grad / EMA target	BYOL, SPR, MR.Q
Distribution matching	**SIGReg** (LeJEPA, LeWM)

1.1 Mechanisms

See arXiv and Theorem 1.4.

Lemma 1.4. $\int_0^1 x \, \mathrm{d}x = \frac{1}{2}$. Code: `latexmk -pvc note.tex`.

A Proofs

Theorem 1.2 (Cramér–Wold). A random vector $\mathbf{z}$ is $\mathcal{N}(\mathbf{0}, \mathbf{I})$ iff $\mathbf{u}^\top \mathbf{z} \sim \mathcal{N}(0, 1)$ for every unit $\mathbf{u}$.

Proof of Theorem 1.2. Characteristic functions agree on every line through the origin. □

List of Theorems

1.1 Definition (Collapse) . . . 1
1.2 Theorem (Cramér–Wold) . . . 1
1.3 Remark . . . 1
1.4 Lemma . . . 2

Notes for the ELSA Lab discussion, October 2026.

[1] With `layout=margin` this note sits in the margin.